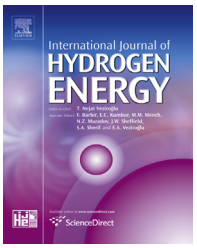
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Shock tube study on ignition delay of hydrogen and evaluation of various kinetic models

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ABSTRACT

H_2/O_2 chemistries are of great crucial for all hydrocarbon and syngas oxidations. Recent study showed that presence of residual H-atom in shock tubes may heavily influences experimental determination of ignition delay times in H_2/O_2 mixtures. Although numerous studies of hydrogen ignition delay times were reported, none of them discussed the residual H-atom in their studies which may impact the accuracy of their measurements. In this study, a systematic measurement related to the ignition delay times were conducted for H_2/O_2 mixtures diluted with argon over the temperatures from 850 K to 1500 K, the pressures from 1.2 to 16.0 atm, and the equivalence ratios of 0.5, 1.0 and 2.0. Special attentions were paid to eliminate the influence of residual H-atom on ignition delay time and the calculation showed that the residual H-atom in current shock tube shows no obvious effect on ignition delay time, which guarantees the present data more convincing. The measurements showed that the hydrogen mixtures have different global activation energies under different pressures. Thus, correlations were derived under different pressures based on all experimental data, which fit fairly well with the experimental data. A couple of detailed kinetic models from literature were validated against the present experimental results. The result shows that the mechanism of Kéromnès-2013 yields the best agreement with data from shock tube experiments at various pressures up to 16 atm, while Davis-2005, GRI 3.0, SanDiego-2011 and Li-2004 mechanisms present significant poor predictions at higher pressures. This investigation is contrary to the previous studies. Reaction pathway and sensitivity analysis indicated that reaction R10: $H + O_2$ (+M) <=> HO_2 (+M) is of significant important pressure-dependent reaction in controlling of hydrogen oxidation. It is inferred that significant differences of their rate constants among various mechanisms induce the difference of performance on hydrogen ignition predictions.

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Introduction

Studies on clean-alternative fuels have been motivated by environmental and energy supply issues. Among various alternative fuels, hydrogen is regarded as a promising alternative fuel in transportation, heating, and power plants in the future due to some favorable combustion characteristics [1,2]. The main property of hydrogen relevant to its engine compatibility is its high flame speed, resulting in broad burn

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Table 1 – Composition of the test mixtures.

Mixture	ϕ	% H_2	% O_2	% Ar
2	0.5	3.472	3.472	93.056
5	1	5.917	2.959	91.124
8	2	9.132	2.283	88.584

limit and can increasing the fraction of exhaust gas recirculation. In addition, hydrogen is the most abundant element in the universe without any carbon, leading to essentially no carbon related pollution. Moreover, H_2/O_2 chemistries play of important part in every hydrocarbon and oxygenate oxidation mechanism because it contains many important elementary reactions involving H, O, OH, HO_2, H_2O, and H_2O_2 that play significant roles in all stages of hydrocarbon oxidation [3]. Therefore, an accurate and reliable kinetic mechanism for hydrogen that can represent its combustion characteristics over a wide range of conditions is historically pursued.

Numerous hydrogen oxidation mechanisms have been reported in the past decades [4–15], which were motivated by the substantial progress achieved in accurate measurement of the elementary reaction rates and thermodynamic properties. The kinetic mechanisms of Ó Conaire et al. [5] and Li et al. [6] were refined by incorporating the best up-to-date available kinetic and thermos data into the mechanism of Mueller et al. [4]. Davis et al. [7] proposed a detailed H_2/CO mechanism based on H_2/CO sub-mechanism of the GRI-Mech version 3.0 [16] by considering the recent revisions in the rate coefficient of H + O_2 (+M)<=> HO_2 (+M), its third-body efficiencies, and the enthalpy of formation of the OH radical. Zsély et al. [8] refined the H_2/CO sub-mechanism of the Leeds methane oxidation [17] mechanism by incorporating the latest rate constant evaluations [18]. Special attention has been focused in this work [8] to the uncertainties of the calculation caused by the uncertainties in the rate constants. A detailed H_2/CO mechanism with rate parameters for elementary steps in the mechanism for hydrogen and to deletion of a hydrogen initiation step and addition of an initiation step for carbon monoxide was proposed by Saxena et al. [9]. Konnov et al. [11]

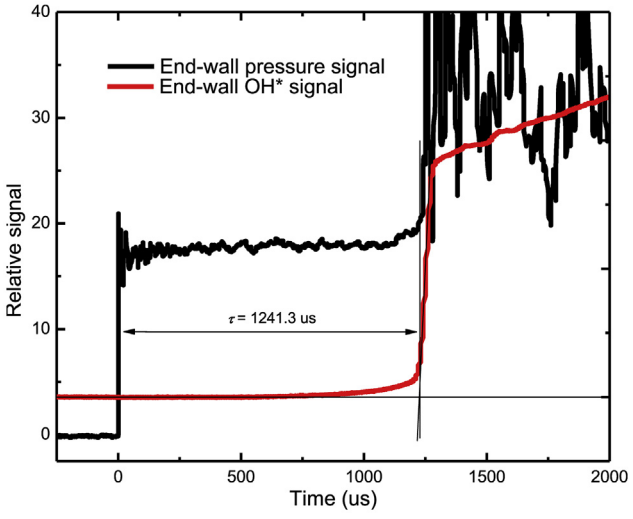


Fig. 1 – Typical end-wall pressure and OH* chemiluminescence measurements the corresponding ignition delay time for a stoichiometric $H_2/O_2/Ar$ mixture at 16.63 atm and 1082.4 K.

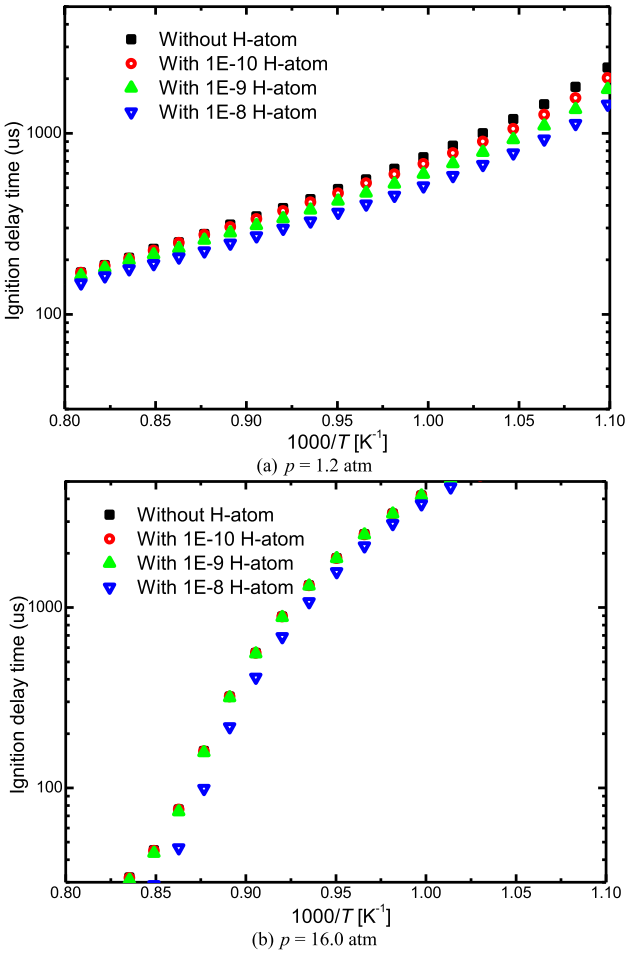


Fig. 2 – The effect of residual H-atom on simulations at $p = 1.2$ and 16.0 atm for the stoichiometric $H_2/O_2/Ar$ mixture using the mechanism of Kéromnès et al. [14].

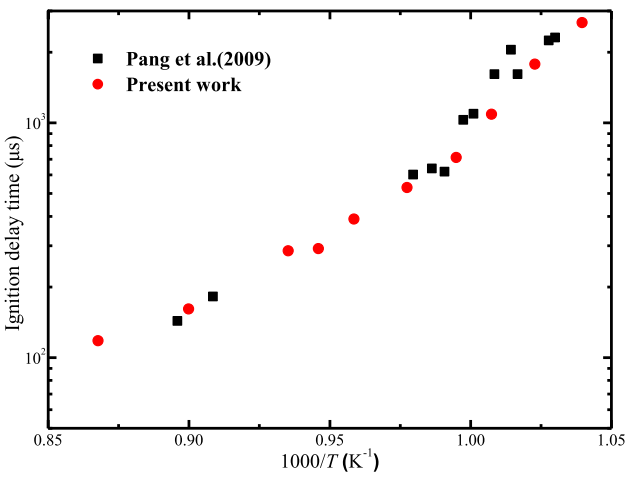


Fig. 3 – Comparison between present measured data and the measured data from reference for 4%H_2/2%O_2/94%Ar at $p = 3.5$ atm and $\phi = 1.0$.

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Table 2 – Correlation parameters for hydrogen mixtures.

p (atm)	T(K)	A	a	Ea(kcal/mol)	R^2
p = 1.2 atm		1.46E-01	−0.080	17.08	0.973
p = 4.0 atm	T > 1039 K	1.15E-02	0.037	20.66	0.968
	T < 1039 K	9.95E-11	0.099	59.01	0.967
p = 10.0 atm	T > 1094 K	9.51E-09	−0.009	51.62	0.957
	T < 1094 K	1.86E-12	−0.313	70.98	0.969
p = 16.0 atm	T > 1142 K	2.20E-09	0.251	56.07	0.976
	T < 1142 K	1.63E-07	−0.331	47.83	0.988

presented a an updated hydrogen combustion mechanism with particular attention paid to different channels of reaction between H atoms and HO_2 radicals, to pressure dependence of the recombination of HO_2 radicals, and to the anomalous rate constant of reaction between OH and HO_2 radicals. Lately, four different updated hydrogen or syngas mechanisms were proposed by Hong et al. [12], burke et al. [13], Kéromnès et al. [14] and Varga et al. [15]. It should be noted that all mechanisms discussed above have been validated against a wide range of experimental conditions and were found to be in good agreement with experimental data from shock tubes, flow reactors, and laminar premixed flames.

Comparisons among different mechanisms and experimental targets are often dedicated to identify the best performance model. Though numerous hydrogen mechanisms have been reported, few studies were touched on comparison of the discrepancy of various hydrogen mechanisms. Ströhle et al. [10] conducted a comparison study among six mechanisms of different research groups with a large set of ignition

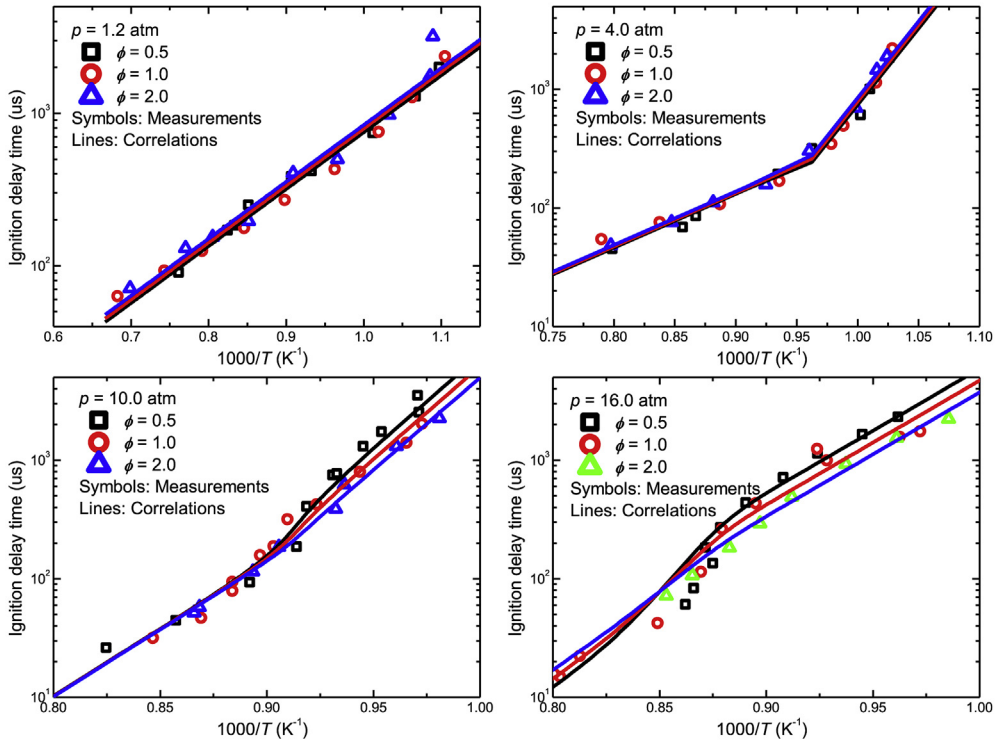


Fig. 4 – Comparison of correlations to all measured data of hydrogen under different equivalence ratios at p = 1.2, 4.0, 10.0, and 16.0 atm.

Table 3 – Details of the hydrogen kinetic mechanism investigated in this study.

NO.	Mechanism ID	Ref.	Update date	Species number	Reactions number
1	GRI3.0	[16]	1999	10	21
2	Ó Conaire-2004	[5]	2004	10	21
3	Li-2004	[6]	2004	11	27
4	Davis-2005	[7]	2005	11	26
5	SaxenaWilliams-2006	[9]	2006	11	21
6	Konnov-2008	[11]	2008	10	33
7	Hong-2011	[12]	2011	10	31
8	SanDiego-2011	[64]	2011	11	21
9	Burke-2012	[13]	2012	10	25
10	Creck-2012	[65]	2012	11	21
11	Kéromnès-2013	[14]	2013	12	33
12	Varga-2015	[15]	2015	12	30

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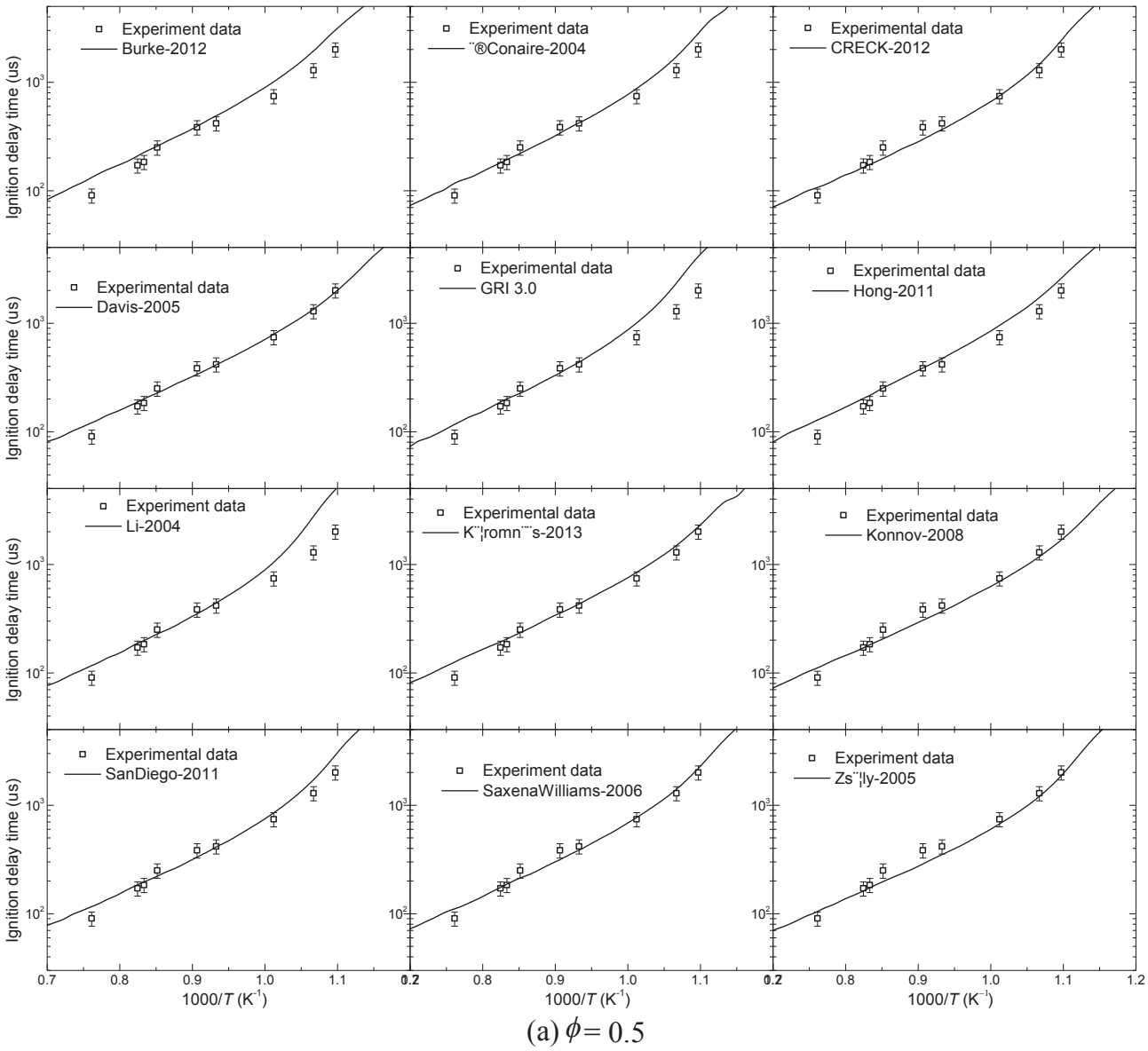
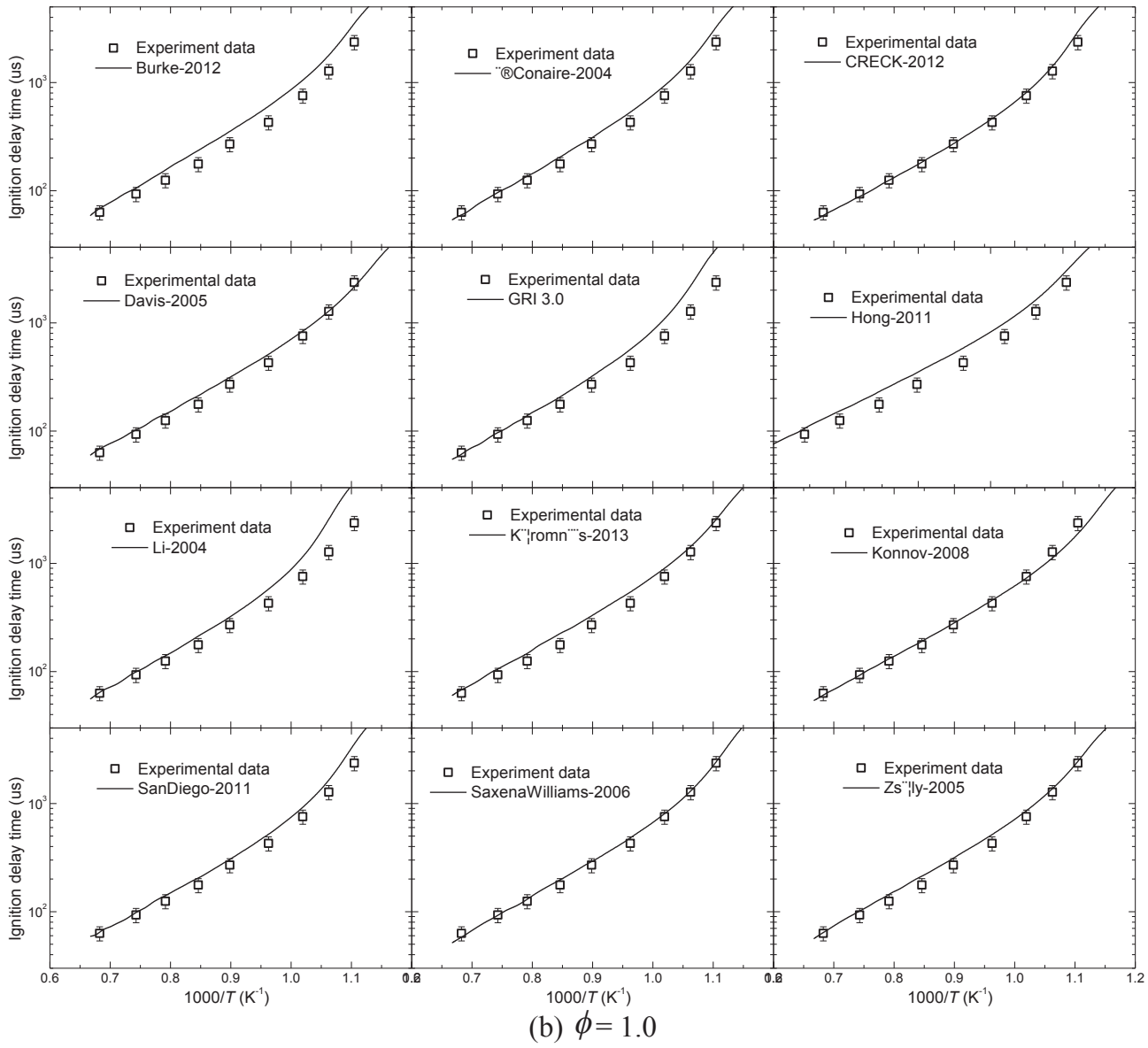


Fig. 5 – Measured and calculated ignition delay time for H_2 mixture at $p = 1.2$ atm.

delay times and flame speeds. This study suggested that the mechanism of Li et al. [6] is best applicable for the prediction of H_2/O_2 combustion. However, this finding is incongruous with the studies made by Pan et al. [19,20] which indicated that the mechanism of Li et al. [6] over-predicted ignition delay times at the high pressure ($p = 10$ atm). It is noted that the ignition delay times validated against in Ströhle et al. [10] are from different sources, which may influenced the comparison results due to different experimental facilities and definitions of ignition delay time. Recently, the performance of all important hydrogen reaction mechanisms published in the last decade were compared against a large set of experimental data by Olm et al. [21]. This study also showed that the mechanism of Li et al. [6] over-predicted ignition delay times at high pressures. The significant discrepancy among these studies highlights that further study and accurate experimental data over wide range of conditions are worthwhile.

Ignition delay times are widely used for the calibration of combustion mechanisms at intermediate to high temperatures. Shock tubes are suitable to accurately determine ignition delay times, as well as to investigate the chemical kinetics of combustion systems. However, the presence of non-ideal effects in this type of experiments [22], especially the interaction of the shock wave and the boundary layer, may influence the measurements. Numerous shock tube experimental ignition delay times of H_2/O_2 are reported in the literature [23–47]. The criteria for defining ignition delay time in these experiments are not identical. In addition, the experimental conditions in these studies are found to be limit to the low pressure or a certain conditions. Moreover, none of this study discussed the influence of residual H-atom on their measurements. All these together make those studies difficult to systematically test the available mechanisms against ignition delay times over a wide range condition.

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(b) $\phi = 1.0$

Fig. 5 – (*continued*).

Thus, the primary goal of this study is to provide reliable ignition delay times of hydrogen using a well-developed shock tube and to extend the available experimental database of hydrogen to a wider range of equivalence ratios and pressures. The effect of residual H-atom on ignition delay time is investigated. The performances of all important hydrogen kinetic mechanisms published in the last decade are compared against the measured ignition delay times. Finally, reaction pathway and sensitivity analysis were conduct to interpret experimental observations and improve several mechanisms.

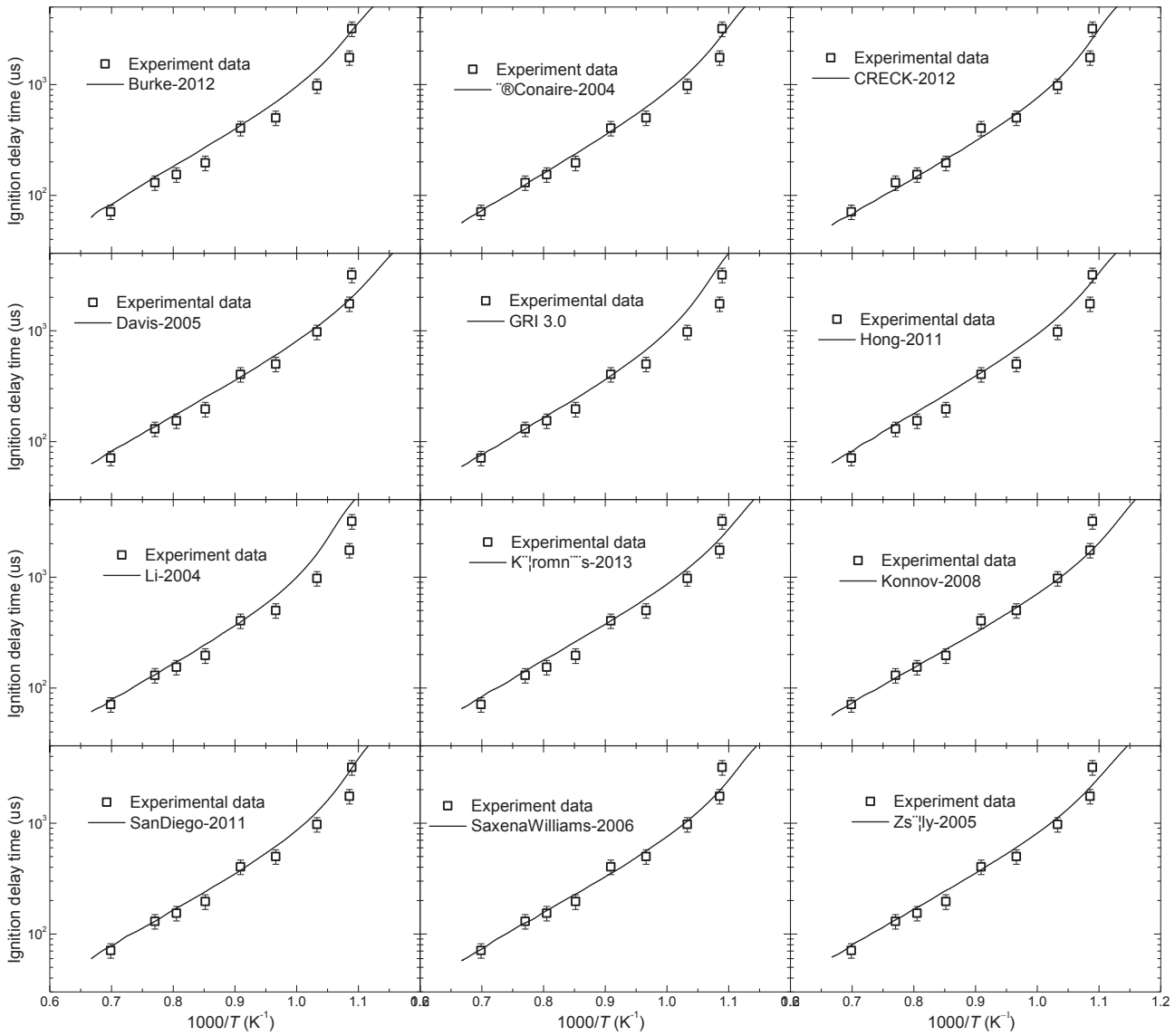
Experimental approach

Experimental setup

The shock tube used in this study has been well characterized and described in former studies, which has been introduced extensively [19,48,49]. Briefly, the main tube with a 11.5 cm diameter is divided into a 4 m long driver section and 5.3 m long driven section by a 0.06 m long connecting flange with polyethylene terephthalate (PET) diaphragms on both sides. To ultimately eliminate the influence of the H-residual impurities in shock tubes [50], the whole shock tube was washed by high-purity argon and whole shock tube was evacuated to pressure below 10^{-6} bar before each experiment. And then after each experiment, whole shock tube was evacuated to pressure below 10^{-6} bar again to expel exhaust gas. this procedure ultimately ensure the accuracy of the measured data. The combustible mixtures in Table 1 are prepared in a 128 L stainless-steel mixing tank. The purities of hydrogen, oxygen and argon are all 99.999%.

Three time counters (Fluke PM 6690) activated by four piezoelectric pressure transducers (PCB 113B26) with nitrile-butadiene rubber to prevent thermal shock from affecting the pressure measurements are employed to measure the

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(c) $\phi = 2.0$

Fig. 5 – *(continued)*.

incident shock velocities. The end-wall incident shock velocity is then determined by linear extrapolation of these three incident shock velocities to the end-wall. The GASEQ [51] developed by the Combustion Chemistry Center in National University of Ireland was used to calculate temperatures (T_5) and the uncertainty of the temperatures will be discussed later. A piezoelectric pressure transducer with acceleration compensation (PCB 113B03) installed at the end-wall was applied to record the pressures (p_5) behind reflected shock waves. The OH* emission selected by a wavelength 307 nm filter is recorded by a photomultiplier (Hamamatsu CR 131) mounted at the end-wall. The definition of ignition delay time is the time spacing between the arrival of reflected shock waves at end-wall and the linear extrapolation of the steepest rise of OH* emission signal to zero level, as shown in Fig. 1.

In this study, Senkin code [52] in the Chemkin II package [53] in conjunction with the SENKIN/VTIM approach [54] to consider the non-ideal effect on calculations was employed to calculate ignition delay time of hydrogen. Simulated ignition delay time is defined as consistent with that of experiment. Rate of pressure rise of 4%/ms based on experiment in our previous work was considered [19,20,55–58].

Analysis of residual H-atoms

The presence of residual H-atom in shock tubes may heavily influences experimental determination of ignition delay times in H_2/O_2 mixtures [50]. Therefore, to eliminate the influence of the H-residual impurities in shock tubes, the whole shock tube was washed by high-purity argon and was evacuated to pressure below 10^{-6} bar before and after each experiment which means residual gas and the residual H-atom is negligible compared to 50–500 torr fuel mixtures. Though done this, it is still essential to calculate the maximum amount of residual H-atom and its influence. Assuming that 10% the hydrogen in the last experiment were converted into H-atoms

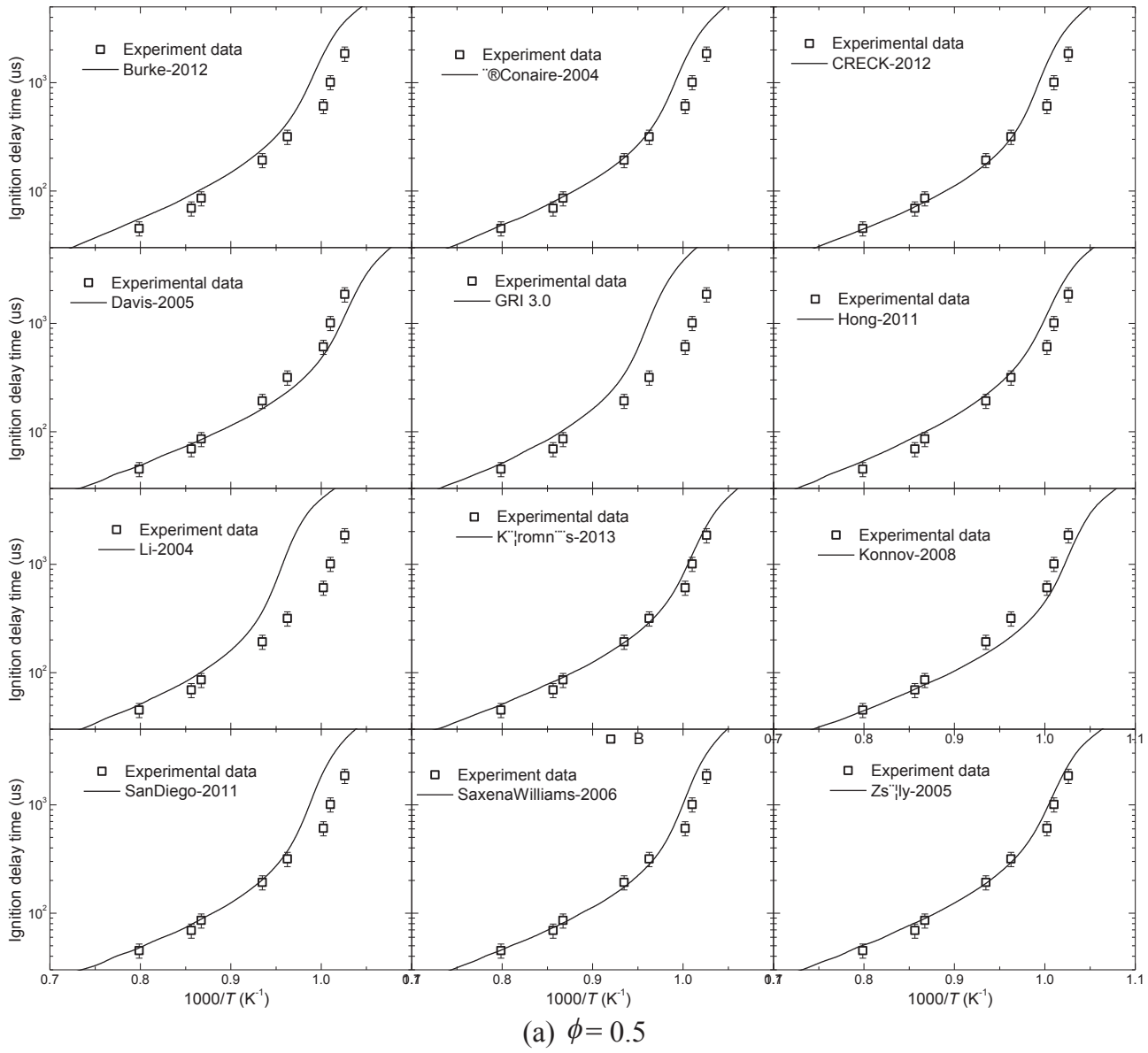


Fig. 6 – Measured and calculated ignition delay time for H_2 mixture at $p = 4.0$ atm.

residual in the exhaust gas, the residual H-atoms in the following experiment can be calculated through following equations:

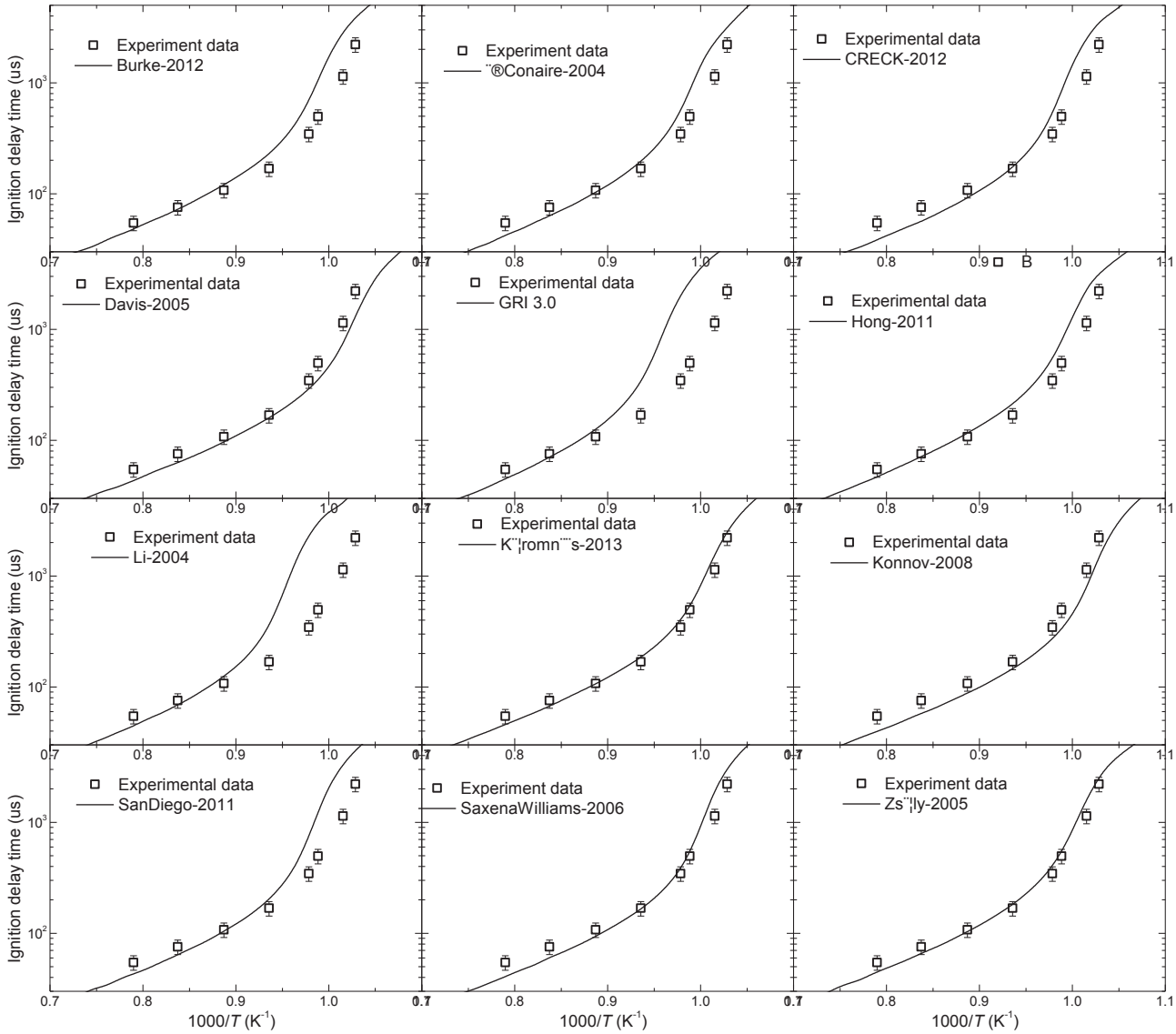
$$X_{1,H-atoms} = 2 \times X_{H2} \times 0.1 \quad (1)$$

$$X_{2,H-atoms} = X_{1,H-atoms} \times 0.1/1 \times 10^5 \quad (2)$$

$$X_{H-atoms} = X_{2,H-atoms} \times 0.1/p_1 \quad (3)$$

X_{H2} is the hydrogen mole fraction in the combustible mixture in the last experiment. $X_{1,H-atoms}$, $X_{2,H-atoms}$ and $X_{H-atoms}$ are the H-atoms mole fraction in the exhaust, after washing by argon and in the following experiment, respectively. p_1 is the pressure of the combustible mixture in the following experiment. The ranges of the hydrogen mole fraction are from 0.035 to 0.091 as provided in Table 1, and the pressure ranges of the combustible mixture are from 50 to 500 torr. Based on above discussion, the maximum mole fraction of the residual H-atoms in present study should be much lower than 4×10^{-11}. Fig. 2 gives the influence of the residual H-atom using the mechanism of Kéromnès et al. [14]. When the concentration of residual H-atom is lower than 10^{-10}, it will have a negligible effect on ignition delay times at both pressures. Moreover, the residual H-atom has a more significant effect on ignition delay time at lower pressure than that at higher pressure, indicating that more attention should be paid when conducting measurement at lower pressure. Considering the maximum of residual H-atom in our shock tube was scaled to be 4×10^{-11} in mole fraction, the influence of residual H-atom therefore can be ignored.

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(b) $\phi = 1.0$

Fig. 6 – (continued).

Analysis of measurement uncertainty

Two categories of measurement uncertainties were considered in this study. The first category is the inaccuracies that affect post reflected shock temperature (T_5). The other one is the uncertainties caused by the presence of residual H-atom. The dominating factor affecting T_5 is the uncertainty of estimation of the incident shock wave velocity. In this study, three time counters (Fluke PM6690) with an uncertainty of 1 μs were used to record the time intervals between the adjacent transducers (PCB 113B26). Specifically, the uncertainty of the calculated incident shock wave velocity is due to the uncertainty in the measured distances between the adjacent transducers which was measured to be 0.02 mm in this study. The uncertainty analysis was made according to the work by Petersen et al. [59]. The standard root-sum-squares (RSS) method was used to calculate the uncertainty in temperature behind the reflected shock wave. Basic expressions used are provided here.

$$T_5 = \frac{T_1[2(\gamma-1)M^2 + (3-\gamma)[(3\gamma-1)M^2 - 2(\gamma-1)]]}{(\gamma+1)^2 M^2} = AM^2 + B + CM^{-2} \tag{4}$$

$$\delta V_s = \sqrt{\left(\frac{1}{\Delta t}\delta_{\Delta z}\right)^2 + \left(\frac{-\Delta z}{\Delta t^2}\delta_{\Delta t}\right)^2} \tag{5}$$

$$\delta T_5 = \frac{\partial T_5}{\partial M}\delta M = \left(2AM - 2CM^{-3}\right)\frac{\delta V_s}{\sqrt{\gamma_1 R T_1}} \tag{6}$$

The distance interval between the adjacent transducers were measured to be 300 mm. Assuming the initial temperature (T_1) is 300 K and gas specific heat ratio (γ_1) is 1.67, then the calculated temperature errors for T_5 = 1000 (M = 1.93) and

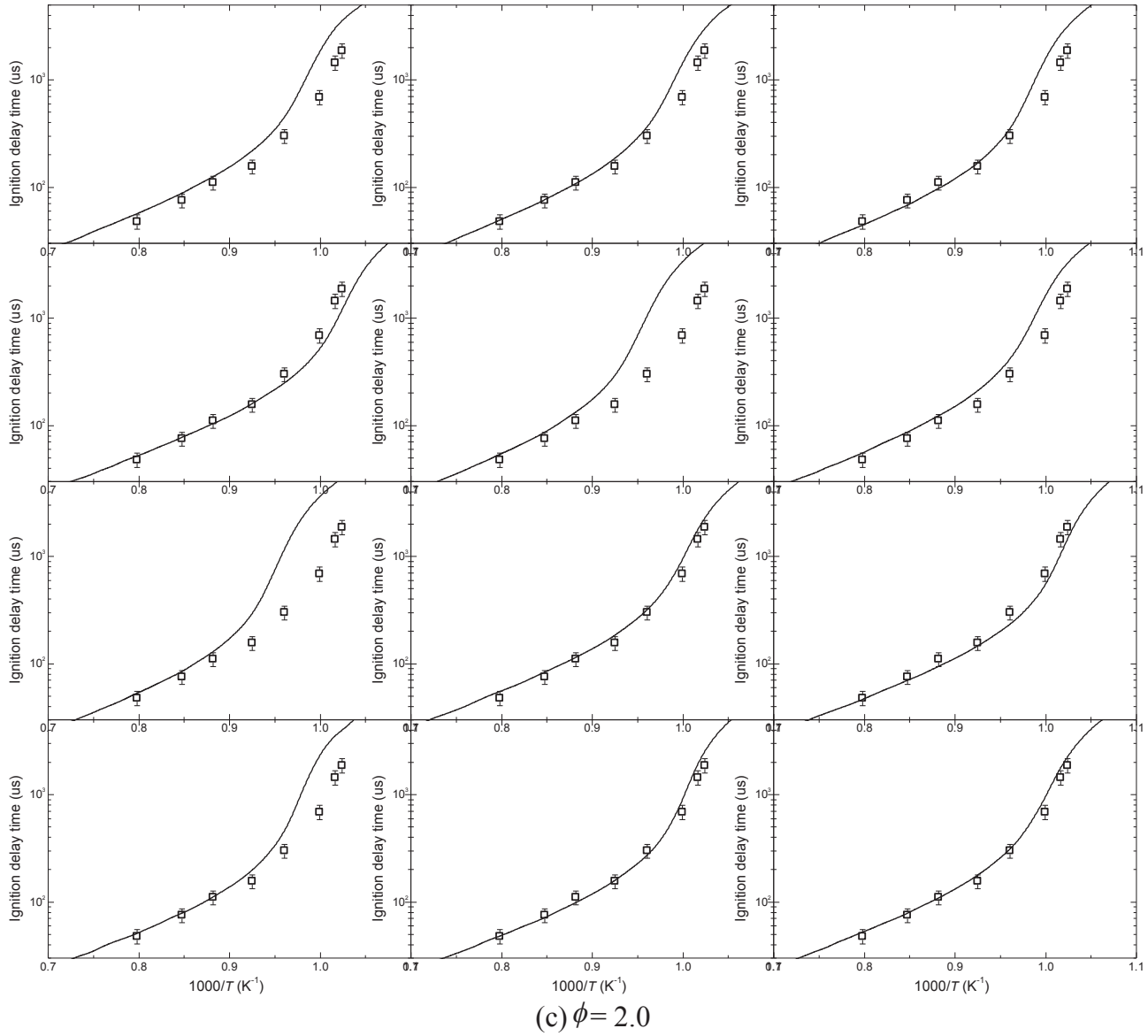


Fig. 6 – *(continued)*.

1200 K (M = 2.17) are about 5 and 7 K, respectively. These small temperature errors lead to less than 8% experimental error of ignition delay time.

The amount of residual H-atom is the governing issue that causes error in measurement of ignition delay time. According the analysis above, the maximum mole fraction of the residual H-atoms in present study was estimated to be 4×10^{-11}, which was calculated to lead to 7% experimental error of ignition delay time. In general, the total experimental error of ignition delay time of this study is less than 15%, which were added to Figs. 5–8.

Results and discussions

In this section, a replicated condition from previous literature for hydrogen was firstly repeated for comparison using our shock tube apparatus. Subsequently, correlations of mixtures were developed based on the measured data. In addition, simulation work was conducted using all important hydrogen reaction mechanisms published in the last decade and finally kinetic studies were conducted to interpret the performance of those mechanisms on ignition delay time.

Comparison with previous data

Though special attentions were paid to measuring ignition delay times, it is still necessary to conduct some comparison studies before high-fidelity validation. Ignition delay behavior of hydrogen has been extensively studied by previous researchers, as discussed in introduction. Recent ignition delay times of hydrogen over a wide temperature range by Pang et al. [45] provides a unique opportunity for the validation of the current measured data, because the Pang et al. [45] study reported the facility-related non-ideal pressure rise as a function of time. Fig. 3 plots the comparison of ignition delay

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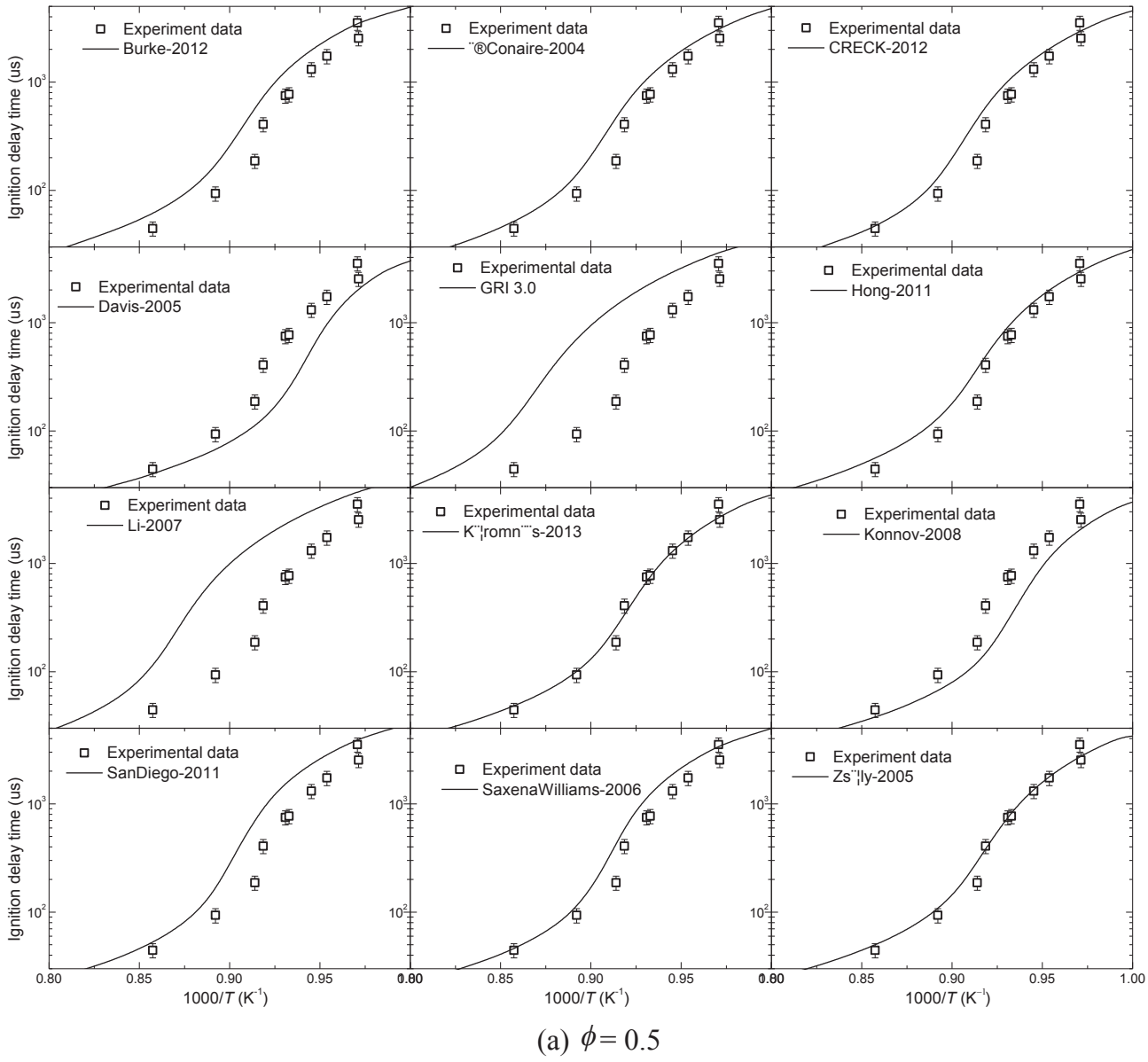


Fig. 7 – Measured and calculated ignition delay time for H_2 mixture at p = 10.0 atm.

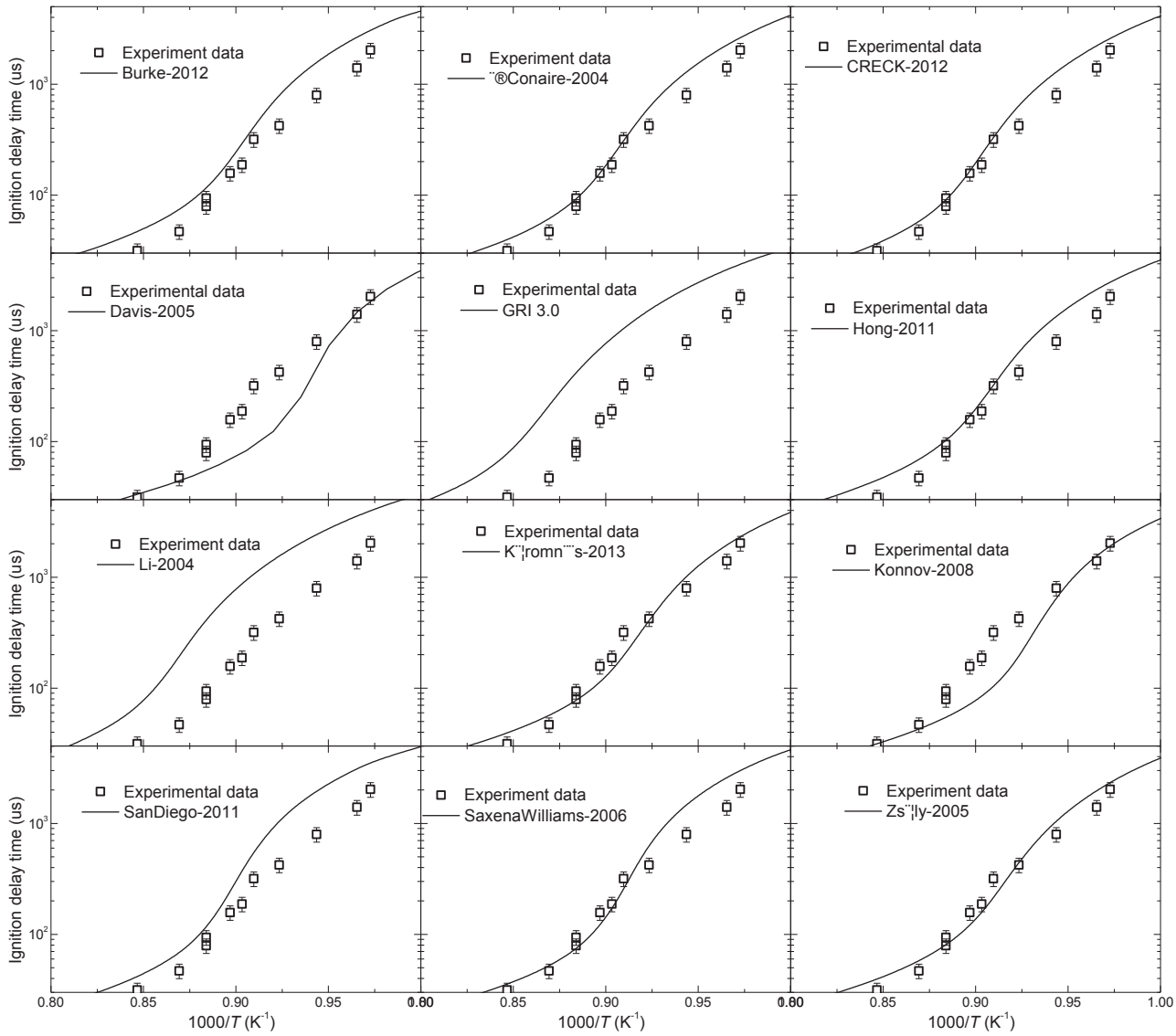
times from Pang et al. [45] and the current measured data. The comparison shows excellent agreement over the entire temperature range. It is noted that considering that the facility-related non-ideal pressure rise is 4%/ms in this work, which is different from Pang et al. [45] (dP^*/dt = 2%/ms), the comparison was only made under ignition delay time shorter than 2 ms. In the study of Pang et al. [45], they found that facility-related non-ideal pressure rise exerts negligible effect when ignition delay time shorter than 2 ms.

Ignition delay time measurements

A systematic measurement related to the ignition delay times of hydrogen was performed behind the reflected shock wave temperatures over 1000–1600 K, pressures of 1.2–16 atm and equivalence ratios of 0.5–2.0. The measurement results showed that the global activation energies of hydrogen ignition delay time are increased with increasing pressure for all three equivalence ratios. In addition, near atmospheric pressure (p = 1.2 atm), the ignition of hydrogen is very similar to the typical hydrocarbon fuels: no distinguished transition of global activation energy is observed as temperature increasing, while an obvious transition of global activation energy is detected with temperature at high pressures (p = 4.0, 10.0 and 16.0 atm). In this study, it is found that the transition points for p = 4.0, 10.0 and 16.0 atm are 1039, 1104 and 1143 K, respectively.

To more directly understand the effect of initial parameters, correlations of ignition delay times for hydrogen were proposed under different pressures. On the basis of previous study and the above discussions, the following format was developed:

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(b) $\phi = 1.0$

Fig. 7 – (continued).

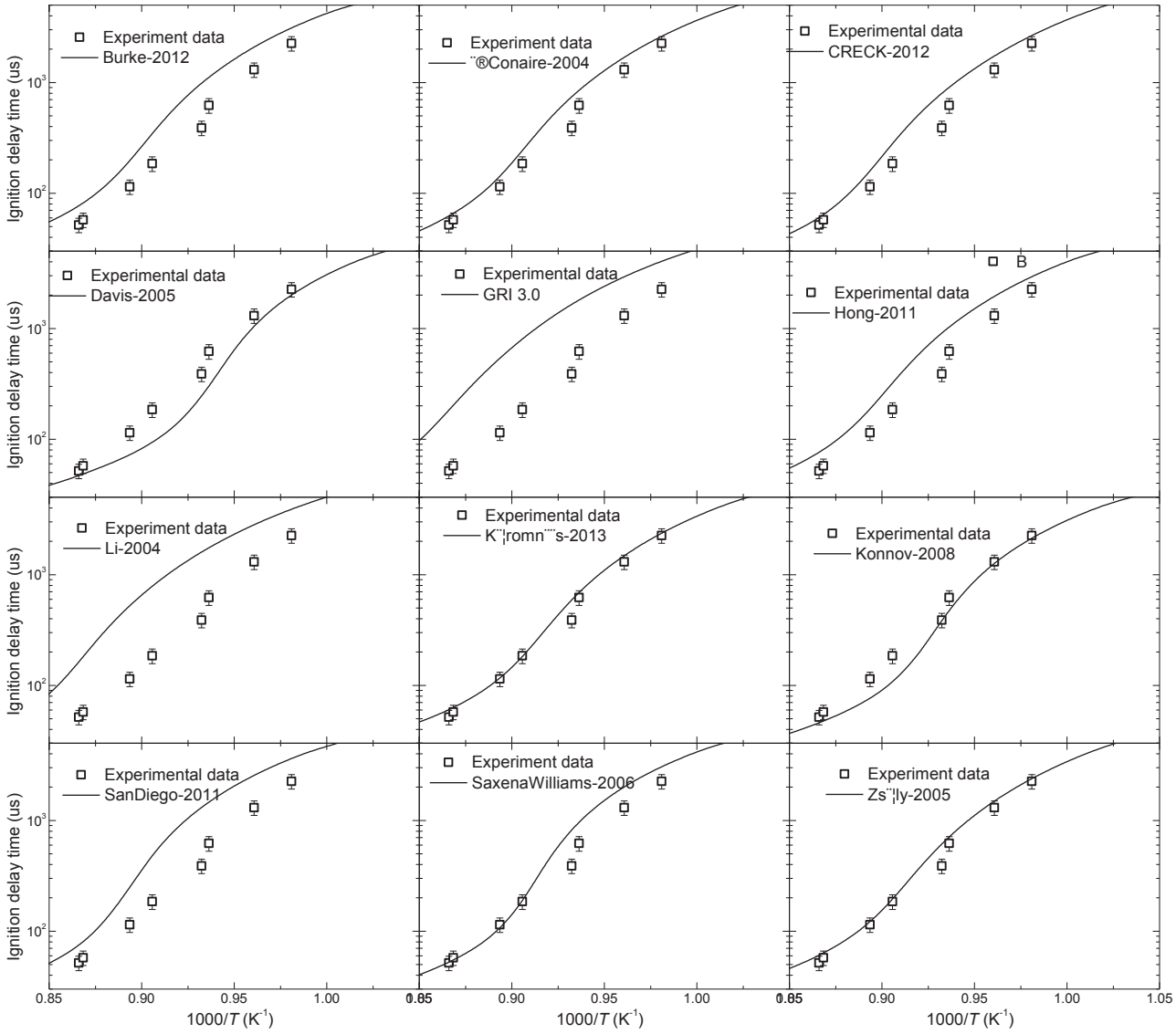
$$\tau = A\phi^{a} \exp(E_a/RT) \quad (7)$$

where τ is ignition delay time in μs, ϕ is the equivalence ratio, T is the temperature in K, $R = 1.986 \times 10^{-3}$ kcal mol^{-1} K^{-1} is the universal gas constant, Ea is the global activation energy in kcal/mol, and A and a are constants. Therefore, correlations were developed through multiple linear regressions on the basis of the measured data. Correlation parameters are summarized in Table 2. It is seen that the global activation energy increases with increasing pressure, indicating that the low-pressure mixture is easier to ignite. Regression coefficients (R^2) of this correlation are higher than 0.95, indicating good regression results. Fig. 4 plots the comparison between the correlated data and all measured data under different equivalence ratio. It can be seen that the correlation data match fairly well with the measured data.

Numerical simulations

Another significant objective of current work is to use our measured hydrogen ignition data to evaluate the performance of some of existing kinetic models over a wide range condition. A large number of detailed mechanisms containing H_2/O_2 kinetics have been reported in the literature. Some of them have been proposed for pure hydrogen oxidation, but most of them are dedicated to hydrocarbons oxidation including sub-mechanisms for H_2/O_2 chemistries. Therefore, it is very difficult to test all of them. In this study, only the major mechanisms published since 1999 are considered, which are provided in Table 3. The measured and predicted ignition delay time for H_2 mixture as a function of temperature at various pressures from the atmospheric pressure to the engine relevant pressure and various equivalence ratios from fuel-lean to fuel-rich is plotted in Figs. 5–8.

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(c) $\phi = 2.0$

Fig. 7 – (*continued*).

Predictions by all mechanisms agree quite well with the measured ignition delay times over the entire temperature, $p = 1.2$ atm and $\phi = 0.5$, 1.0, 2.0, as shown in Fig. 5, indicating that all these reaction mechanisms are capable to predicting hydrogen ignition around atmospheric pressure. However, as pressure increased, obvious discrepancy among the calculations and experimental data is presented. For $p = 4.0$ atm, GRI 3.0 and Li-2004 mechanisms significantly over-predict and Burke-2012 mechanism slightly over-predict the measured data for all the three equivalence ratios at lower temperature, as shown in Fig. 6. The rest of them to some extent match fairly well with the measured data over the investigated temperature range. As pressure continually increase to $p = 10$ atm, the performance of these mechanisms shows significant discrepancy, as shown in Fig. 7. Specifically, GRI 3.0 and Li-2004 mechanisms still give a significant over-prediction for all the three equivalence ratios over the entire temperature range. SanDiego-2011 and Burke-2012 moderately and slightly over-predict the measured data for all the three equivalence ratios, respectively. Davis-2005 and Konnov-2008 mechanisms generally under-predict the measured ignition delay times. Ó Conaire-2004, CRECK-2012, Hong-2011, Kéromnès-2013 and Varga-2015 mechanisms yield relatively satisfactory agreement. For $p = 16$ atm, as shown in Fig. 8, only four mechanisms, Ó Conaire-2004, CRECK-2012, Kéromnès-2013 and Varga-2015 mechanisms, can predict well the measure data. GRI 3.0 and Li-2004 mechanism significantly over-predict the measured ignition delay time, while Davis-2005 under-predicts the measured ignition delay times. In addition, Burke-2012, Hong-2011 and SaxenaWilliams-2006 show slight over-prediction.

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In general, all the studied mechanisms are capable to predicting hydrogen ignition around atmospheric pressure, but only four mechanisms: Ó Conaire-2004, CRECK-2012, Kéromnès-2013 and Varga-2015 can predict well the measure data over the investigated conditions. Noticeably, three generally accepted mechanisms: GRI 3.0, SanDiego-2011 and Li-2004, which are widely used to simulate hydrogen combustion or as a base mechanism to construct higher-carbon mechanisms due to the hierarchical structure, give significant over-prediction on the measured data when pressure is above atmospheric pressure. Therefore, special attentions are paid to illuminate the potential reasons for the poor predictions by Davis-2005, GRI 3.0, SanDiego-2011 and Li-2004 mechanisms.

Chemical analysis

Poor performance model improvement

As shown in Figs. 5–8, all of the mechanisms investigated in this study give excellent prediction on ignition delay time at $p = 1.2$ atm. However, the mechanisms of Davis-2005, GRI 3.0, SanDiego-2011 and Li-2004 yield unsatisfactory performance for hydrogen above atmospheric pressure, especially at $p = 10$ and 16.0 atm. Considering the Kéromnès-2013 mechanism well predicts the measured data under all studied conditions and newly updated, thus was served as the core model to conduct the kinetic analysis in the following. Reaction pathway analysis of the above mentioned five models were conducted at $T = 1100$ K, $p = 16$ atm, and $\phi = 1.0$ to ascertain the controlling steps. As shown in Fig. 9, H-abstraction by O and OH radicals are the dominated pathways consuming hydrogen in all these five models. Additionally, all these mechanisms seem to have a very similar portion for hydrogen consumption pathways at current condition and the percentage of hydrogen molecules consumed by O and OH radicals are around 22% and 78%, respectively. Subsequently, the generated H radicals are primarily consumed by three paths: R1: $H + O_2 <=> O + OH$, R4: $O + OH + M <=> H_2O + M$, and R10: $H + O_2 (+M) <=> HO_2 (+M)$. It noted that these three reactions

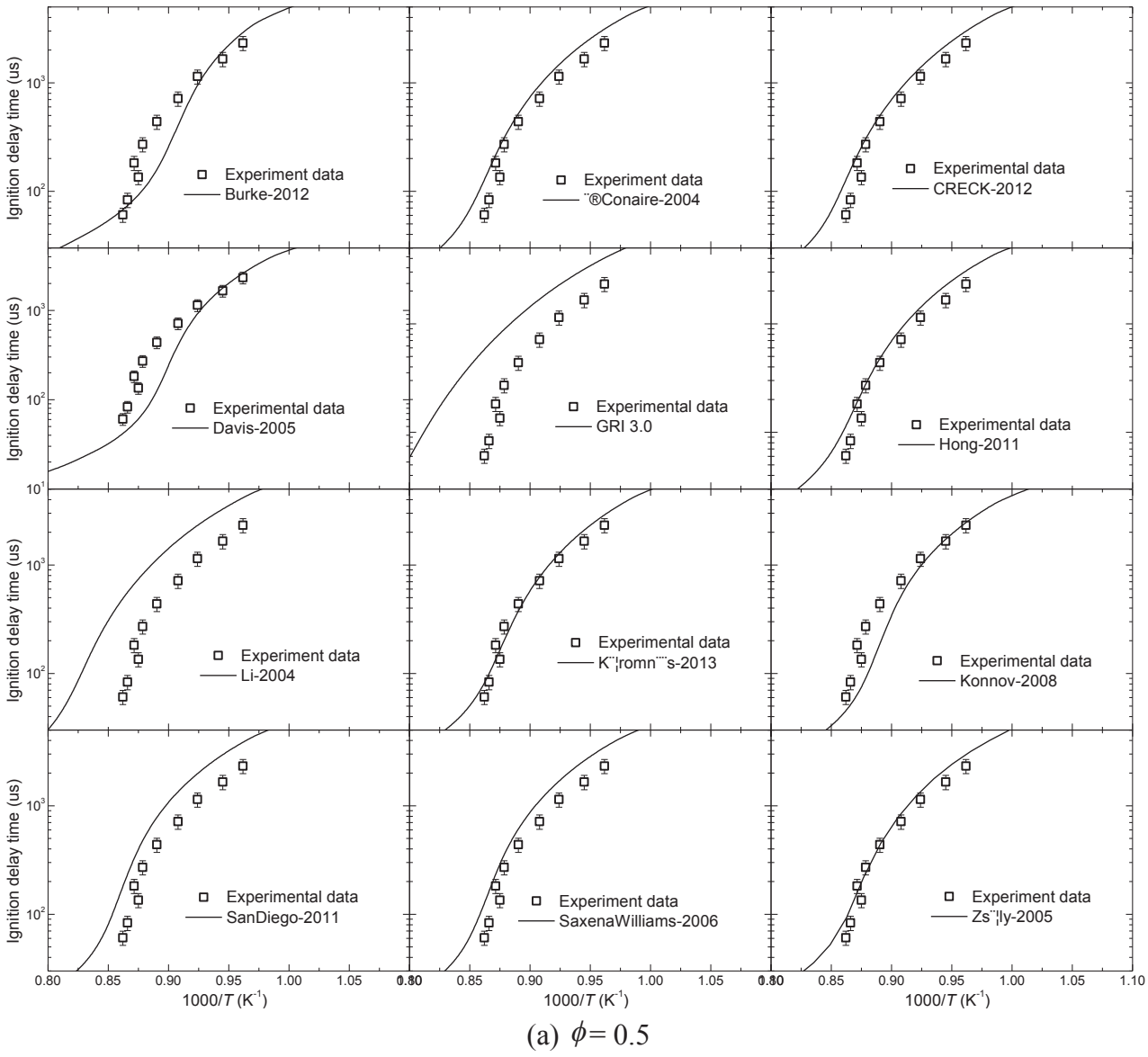
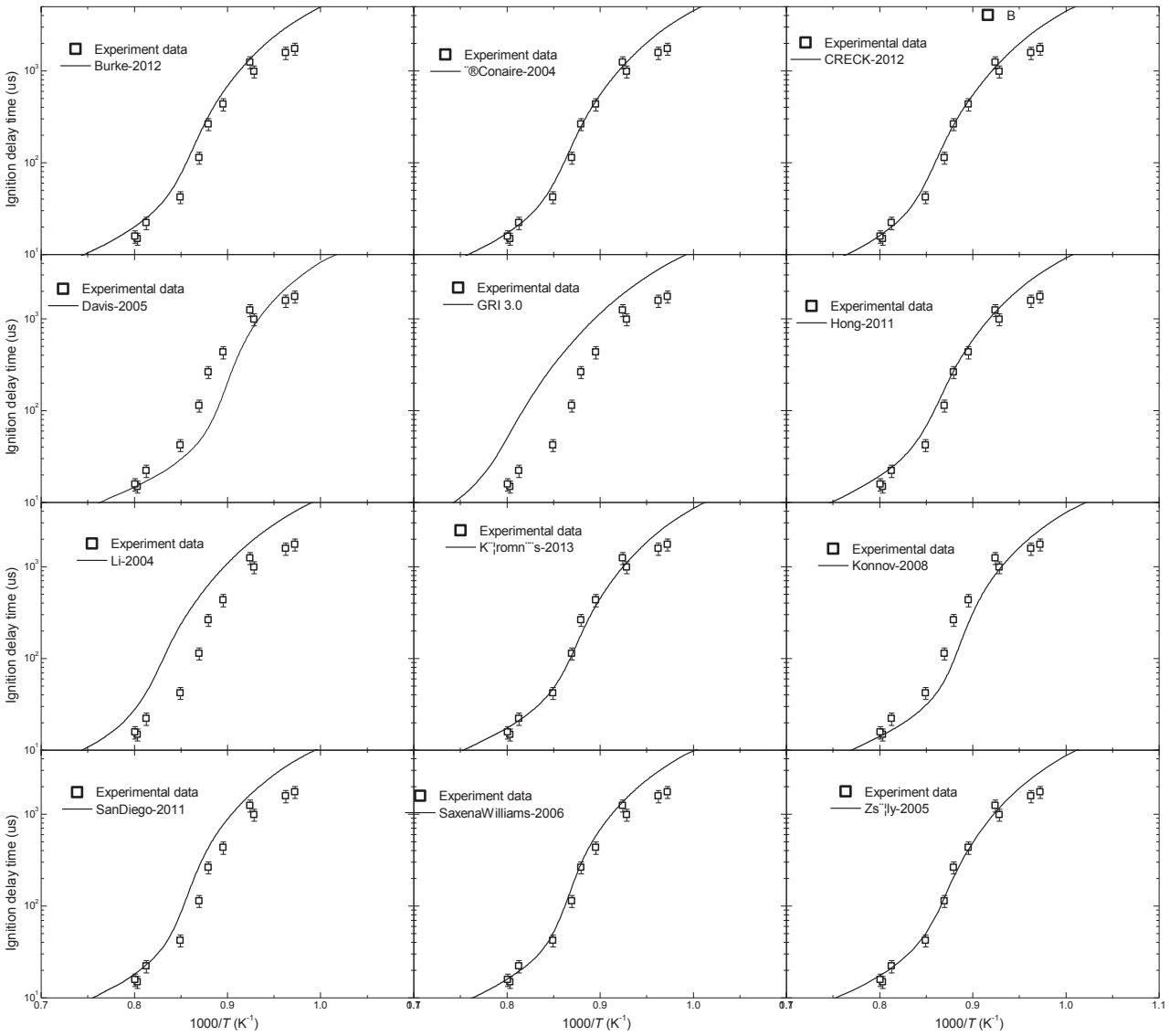


Fig. 8 – Measured and calculated ignition delay time for H_2 mixture at $p = 16.0$ atm.

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(b) $\phi = 1.0$

Fig. 8 – (*continued*).

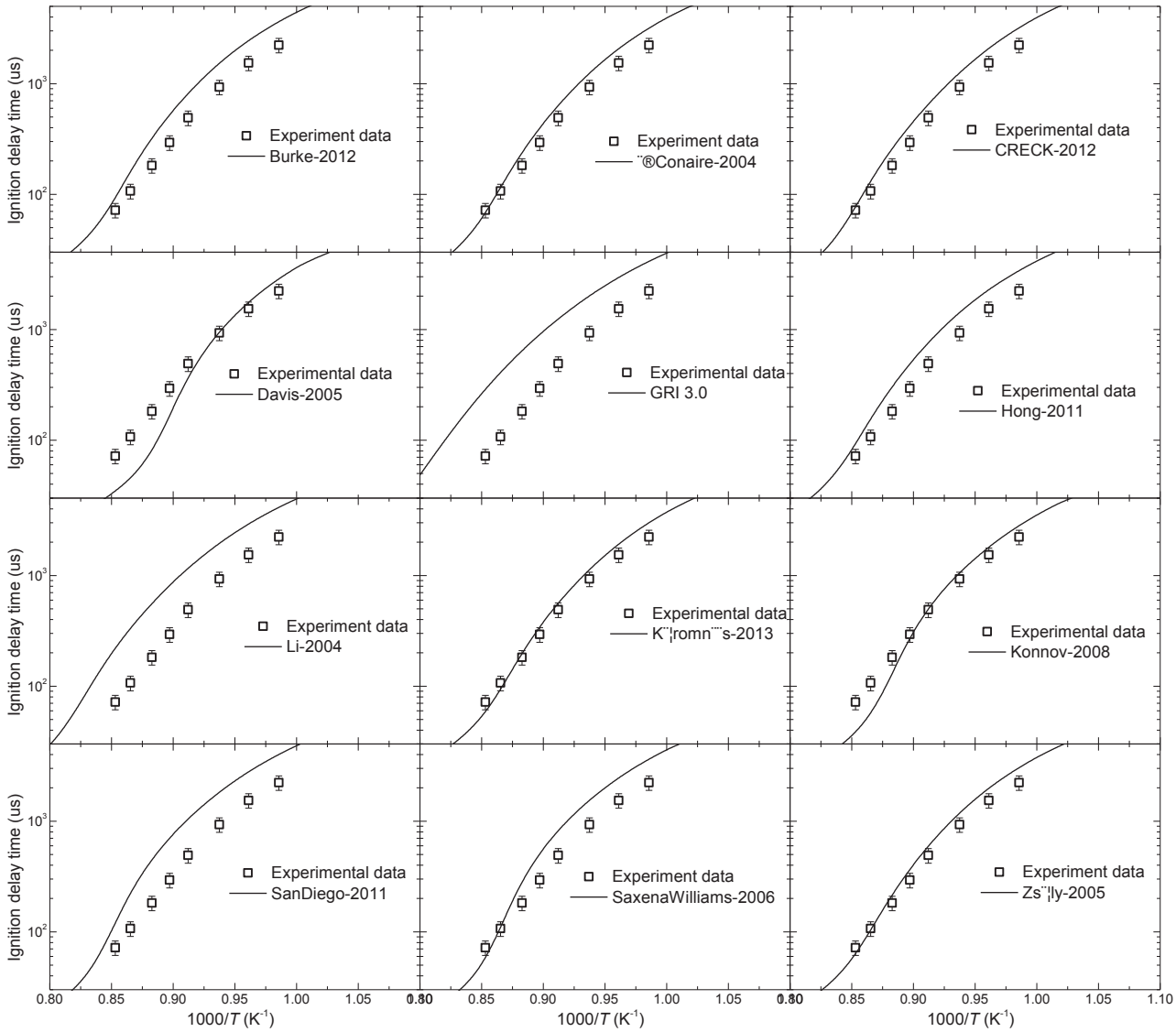
compete H radicals among each other. Moreover, it is found that the different mechanisms have an obvious different portion for H radicals' consumption pathways at current condition. Specifically, the mechanisms that over-predict the measured ignition delay times at $p = 16$ atm have a much higher reaction rate of R10 and a lower reaction rate of R4 and R1 than that of romnès-2013 mechanism. For example, 26.9% of H radicals are consumed by reaction R10 in Li-2004 mechanism, which is about two times than Kéromnès-2013 mechanism. Davis-2005 mechanism which under-predict the measured ignition delay times at current condition, has significant lower reaction rate of R10 than Kéromnès-2013 mechanism. In Kéromnès-2013 mechanism, H radicals consumed by R1 is 15.6%, while in Davis-2005 mechanism is 9%. In addition, Davis-2005 mechanism has a much higher and slightly lower reaction rate of R4 and R10 than those of Kéromnès-2013 mechanism. So it is concluded that the different reaction rate of consuming H radicals among these mechanisms leads to the different prediction results.

To further identify the reactions that responsible for the poor prediction for the mechanisms, sensitivity analysis was performed using Kéromnès-2013 mechanism at the condition of $T = 1100$ K, $p = 16.0$ atm, and $\phi = 1.0$. Definition of sensitivity analysis parameter can be found in numerous studies [19,20] and given as follow:

$$S_i = \frac{\tau(2k_i) - \tau(0.5k_i)}{1.5\tau(k_i)} \tag{8}$$

where τ is the ignition delay time, and S_i and k_i are the normalized sensitivity index and rate constant of the ith

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(c) $\phi = 2.0$

Fig. 8 – (continued).

reaction. A negative coefficient indicates the promoting effect on the overall reactivity and vice versa. Fig. 10 provides the 10 most sensitive reactions. Associating with reaction pathway analysis, it is found that the chain branching reaction R1 is the most promoting reaction, and R10 is the most important inhabiting reactions. Therefore, Davis-2005 mechanism which has highest rate of R1 gives shortest ignition delay time, while GRI 3.0, SanDiego-2011, and Li-2004 whom have higher reaction rate of R10, give longer calculated ignition delay times than Kéromnès-2013 mechanism. Considering all these mechanisms are performance quite well at $p = 1.2$ atm, therefore, it is also inferred that the poor prediction by Davis-2005, GRI 3.0, SanDiego-2011 and Li-2004 mechanisms is more likely relevance to pressure-dependence reaction R10: $H + O_2(+M) <=> HO_2(+M)$.

As a significant important pressure-dependent reaction, the rate constants of R10 in all these models are written in TROE form except GRI 3.0 mechanism. It is found that rate constants of R10 for those mechanisms come from different sources. The rate constants of R10 in Kéromnès-2013 mechanism are new fits determined by Fernandes et al. [60] and Bates et al. [61]. The rate constants of R10 in Davis-2005, Li -2004 and SanDiego mechanism were based on Troe et al. [62], Cobos et al. [63] and Fernandes et al. [60], respectively, whereas GRI 3.0 mechanism employed a non-TROE fit. Therefore, the rate constants of this reaction from these mechanisms are compared in Fig. 11. It should be noted that in order to be consistent with the experiment, argon was used as the bath gas to calculate reaction rates of R10 in Fig. 11. It is observed that reaction rate of R10 for GRI 3.0, SanDiego-2011,

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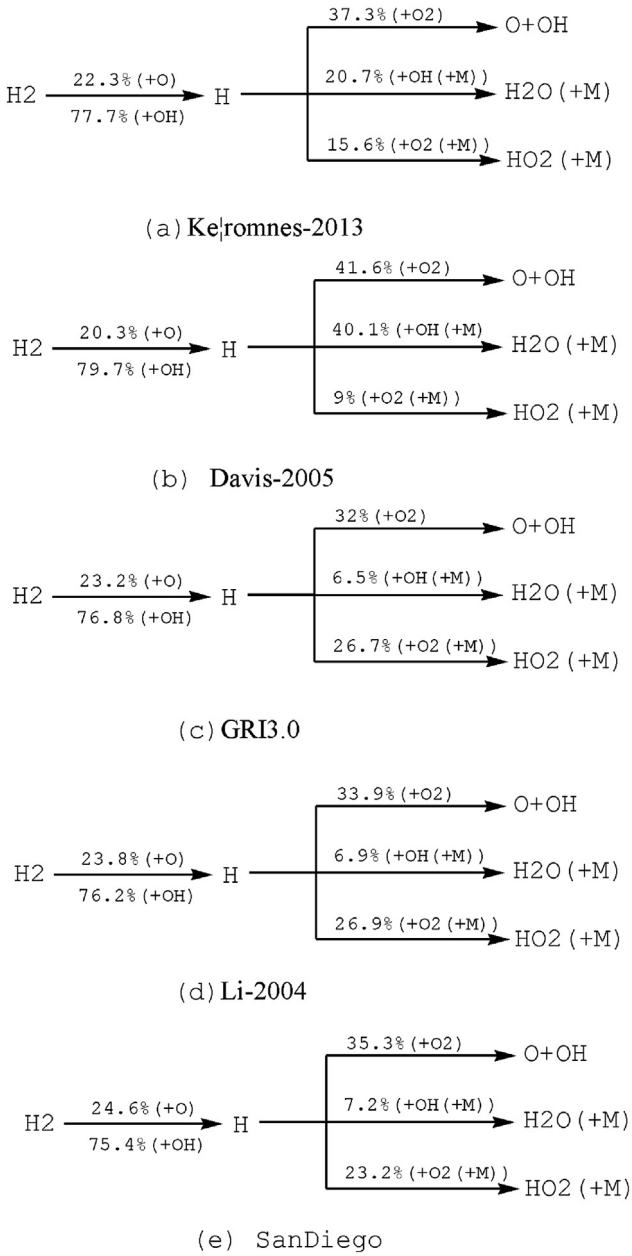


Fig. 9 – Reaction pathway analysis of hydrogen calculated by five mechanisms in shock tube at $T = 1100$ K, $\phi = 1.0$, $p = 16.0$ atm, and 50% fuel conversion.

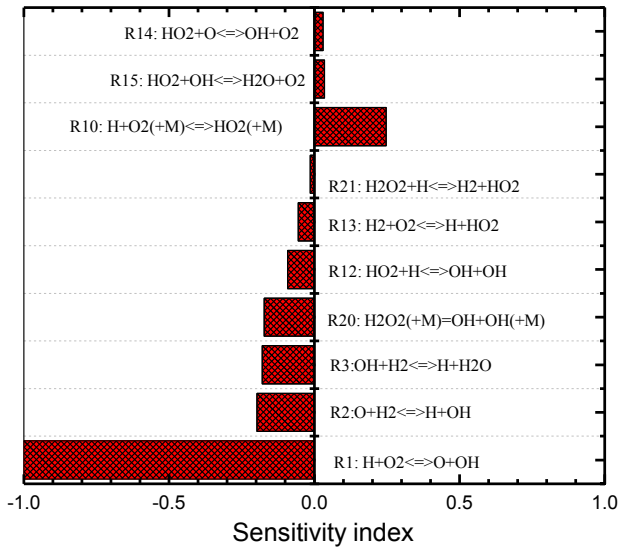


Fig. 10 – Sensitivity analysis of ignition delays calculated by Kéromnès-2013 mechanism at $T = 1100$ K, $p = 16.0$ atm, and $\phi = 1.0$.

Li-2004 are much higher than that in Kéromnès-2013 mechanism, whereas reaction rate of R10 for Davis-2005 mechanism is approximately two times lower than that in Kéromnès-2013 mechanism. To evaluate the effect of these reactions, the reaction constants of R10 in Davis-2005, GRI 3.0, SanDiego-2011 and Li-2004 mechanisms are replaced by the reaction constants of R10 in the Kéromnès-2013 mechanism as shown in Fig. 12. It is observed that the modified Davis-2005, GRI 3.0, SanDiego-2011 and Li-2004 mechanisms yield considerably better agreement with experimental data at all pressures investigated in this study. Additionally, the other reactions with high sensitivity coefficient were replaced with corresponding one in the Kéromnès-2013 mechanism and limited effects on ignition were detected. Therefore, it is inferred that residual higher uncertainty in rate constants of reaction R10 mainly lead to poor prediction for Davis-2005, GRI 3.0, SanDiego-2011 and Li-2004 mechanisms. The rate constants of R10 in Kéromnès-2013 mechanism may not be definitely accurate and thus, further high-level theoretical studies on rate constants of R10 are highly recommended.

Best performance model comparison

As studied above, Ó Conaire-2004, CRECK-2012, Kéromnès-2013 and Varga-2015 models can predict well the measure data over the investigated conditions. Take note that, among the 4 best models, three of them are the latest published and the other one (Ó Conaire-2004 model) is from the same group than Kéromnès-2013 model. So CRECK-2012, Kéromnès-2013 and Varga-2015 models were selected out to further compare and analyze them to explain the differences and make further suggestions for H_2 chemistry improvement. In addition, Burke-2012 model updated recently was also considered in the comparison. Reaction pathway analysis of the above mentioned four models were conducted at $T = 1100$ K, $p = 16$ atm, and $\phi = 1.0$ to ascertain the controlling steps, as shown in Fig. 13. Still, as expected, H-abstraction by O and OH radicals are the dominated pathways consuming hydrogen in all these five models. Although they are well reproduce the measured ignition delay times quite well, the four mechanisms still show to have different portions for hydrogen consumption pathways at current condition and the order for percentage of hydrogen molecules consumed by O is Kéromnès-2013 (22.3%), Vargra-2016 (17.3%), Burke-2012 (16.3%) and Creck-2012 (9.4%). Surprisingly, hydrogen molecules consumed by O in Kéromnès-2013 is two orders that in Creck-2012, indicating these models have significantly discrepancy on H-abstraction reactions, such as R1: $H + O_2 <=> OH + O$ and R2: $O + H_2 <=> OH + H$. In addition, one can observe that H radicals consumed *via* three body collision reaction R10, which exhibits the highest negative impact on

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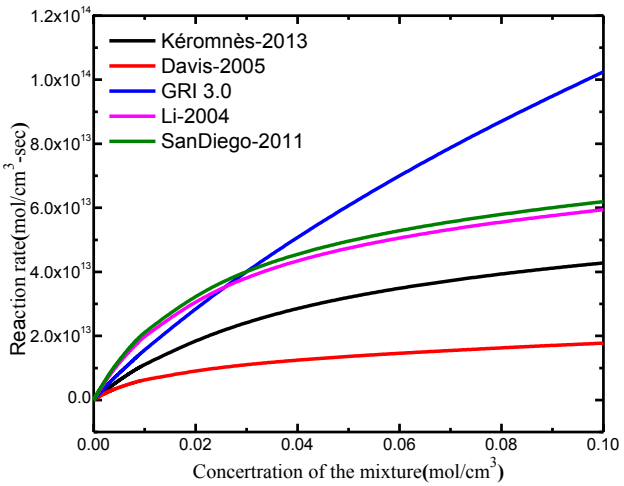


Fig. 11 – Rate constants for reaction H + O_2(+M) <=> HO_2(+M) in Davis-2005, GRI 3.0, SanDiego-2011, Li-2004 and Kéromnès-2013 mechanism.

ignition delay time, also show quite different either. As we all known reaction R1 and R10 are of two most important and ignition sensitive reactions not only in hydrogen oxidation system but also hydrocarbon fuel systems and significantly difference in consumption portion for R1 and R10 in different mechanisms indicates that rate constants for R1 and R10 may be different in different models; therefore, high-level calculations or experimental studies are highly recommended.

Conclusions

Hydrogen chemistries are of significant importance in chemical kinetic modeling of all hydrocarbons and syngas. Therefore, ignition delay times of H_2/O_2/Ar were experimentally measured behind the reflected shock wave over a systematic condition: temperatures of 1000–1600 K, pressures of 1.2–16 atm, and equivalence ratios of 0.5–1.0. The main results are summarized as follows:

(1) The measurements show that global activation energy of hydrogen is increased with increasing pressures and thus, correlations were developed through multiple linear regressions under various pressures.
(2) Measured ignition delay times were evaluated with all important hydrogen reaction mechanisms published in the last decade. Only four mechanisms: Ó Conaire-2004, CRECK-2012, Kéromnès-2013 and Varga-2015 can

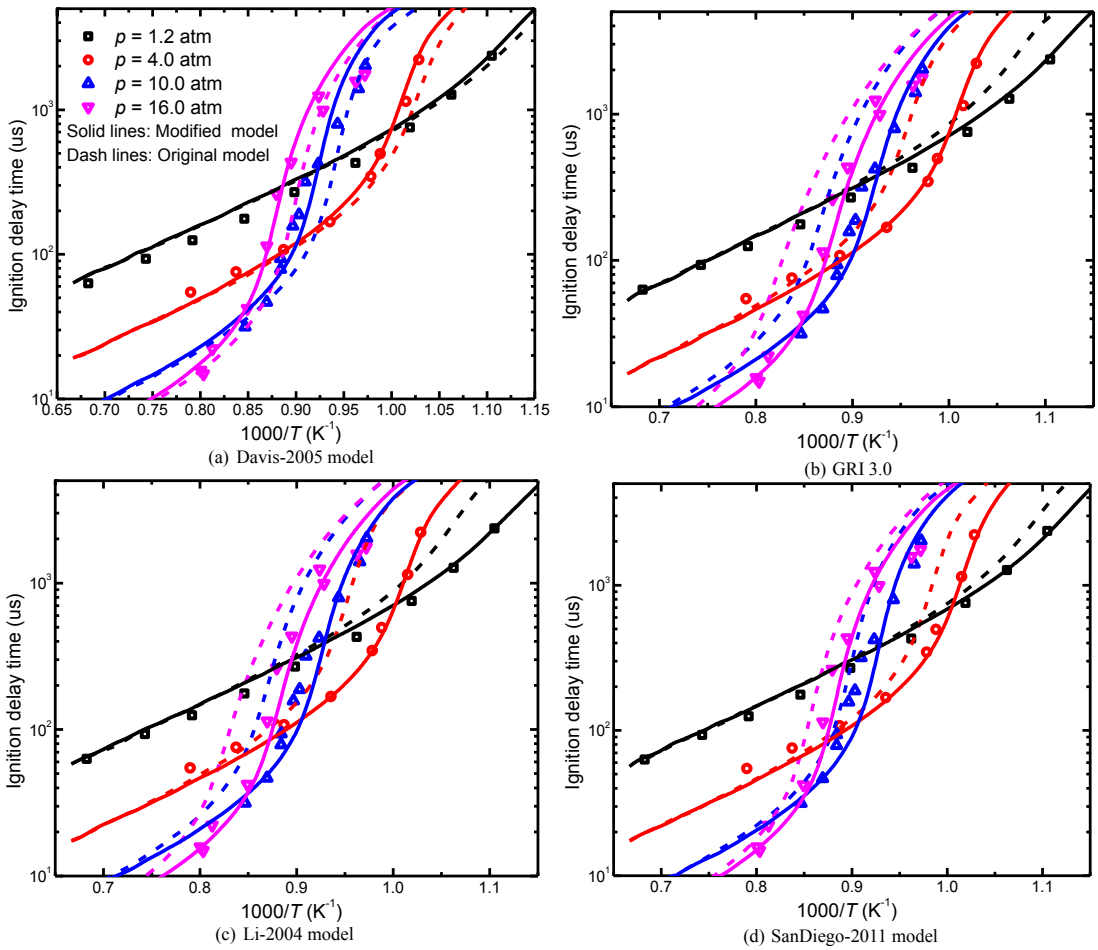


Fig. 12 – Effect of the rate constant of reaction H + O_2(+M) <=> HO_2(+M) on the poor prediction models for the Stoichiometric mixtures.

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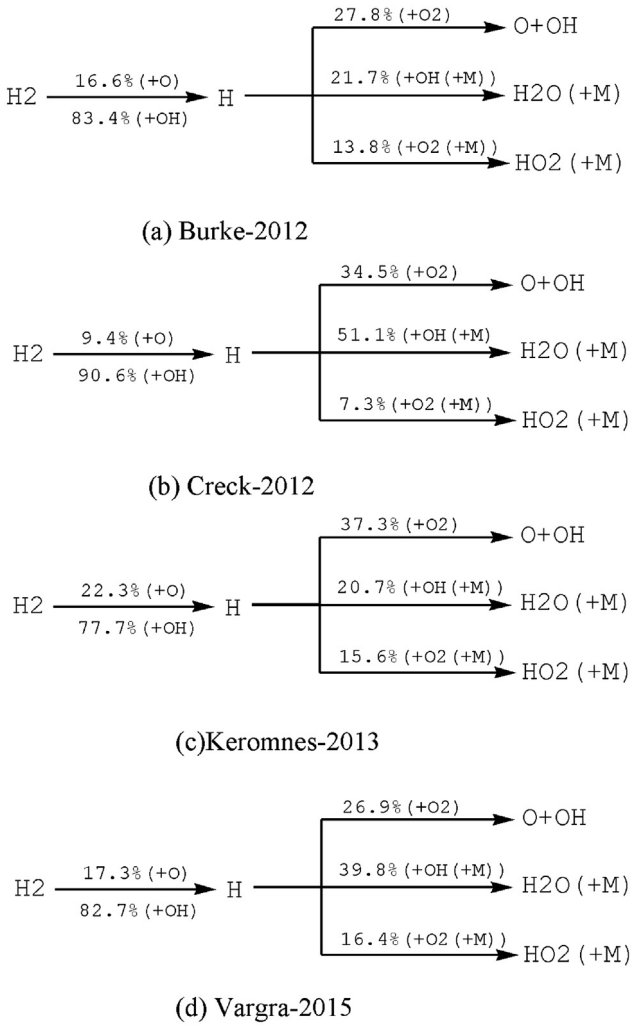


Fig. 13 – Reaction pathway analysis of hydrogen calculated by four better mechanisms in shock tube at $T = 1100$ K, $\phi = 1.0$, $p = 16.0$ atm, and 50% fuel conversion.

predict well the measure data over the investigated conditions. Three generally accepted mechanisms: GRI 3.0, SanDiego-2011 and Li-2004 give significant over-prediction on the measured data when pressure is above atmospheric pressure and another widely used mechanism, Davis-2005 mechanism, significantly under-predict the measured data at $p = 10$ and 16 atm.

(3) Reaction pathway and sensitivity analysis indicated that reaction R10: $H + O_2$ (+M) <=> HO_2 (+M) is of significant important pressure-dependent reaction in controlling of hydrogen ethane oxidation. The significant differences of their reaction rate constants among various mechanisms cause the difference of performance on hydrogen ignition predictions.

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